PRACTICAL-1

Problem Definition:

Study, dissect, and map the complete end-to-end data engineering lifecycle for an enterprise e-commerce platform. Students must trace data states across all five core lifecycle stages: data generation, ingestion, storage, transformation, and serving. The solution requires creating a detailed architectural data flow diagram along with asystem manifesto documenting cross-cutting lifecycle undercurrents (security, observability, and data privacy). This initial structure must be committed to a personal Git repository.

Key Questions:

● **Que / Key Point 1:** How do the technical demands of downstream serving layers directly influence ingestion and storage design decisions during the early generation phases?

Ans:

1. Determine whether batch or real-time ingestion is needed.
2. Decide the data format (structured, semi-structured, unstructured).
3. Influence storage choice (database, data lake, data warehouse).
4. Ensure data quality through validation and cleaning.
5. Improve query performance using partitioning and indexing.
6. Support scalability based on future data usage.

● **Que / Key Point 2:** Differentiate between the broader data lifecycle and the technical data engineering lifecycle within an enterprise.

Ans:

Data Lifecycle	Data Engineering Lifecycle
Covers the complete journey of data.	Focuses on technical processing of data.
Business-oriented.	Technology-oriented.
Includes creation, usage, archival, and deletion.	Includes ingestion, storage, transformation, and serving.
Involves business and compliance teams.	Involves data engineers and architects.

● **Que / Key Point 3:** Identify how the undercurrents of security and data observability should be enforced at each lifecycle boundary.

Ans:

1. Generation: Secure data sources and monitor data collection.
2. Ingestion: Encrypt data transfer and monitor pipeline failures.
3. Storage: Use encryption, access control, and monitor data health.
4. Transformation: Protect processing and track data quality.
5. Serving: Authenticate users and monitor query performance.
6. Maintenance: Apply security updates and continuous monitoring.

Supplementary Problems:

Map out the data engineering lifecycle for a smart city IoT traffic sensor platform, emphasizing challenges related to continuous real-time streaming data generation.

1. Data Generation

- Traffic sensors continuously generate vehicle and road data.
- GPS, cameras, and IoT devices produce real-time streams.
- **Challenge:** High data volume and continuous data generation.

2. Data Ingestion

- Streaming platforms collect sensor data in real time.
- Handle thousands of events per second.
- **Challenge:** Low latency and reliable data transmission.

3. Data Storage

- Store raw data in a data lake and processed data in databases.
- Partition data based on location and time.
- **Challenge:** Scalable storage for massive datasets.

4. Data Processing / Transformation

- Clean, validate, and aggregate sensor data.
- Remove duplicate or missing records.
- **Challenge:** Real-time processing with minimal delay.

5. Data Serving

- Deliver live traffic updates to dashboards and navigation apps.
- Provide APIs for traffic management systems.
- **Challenge:** Fast query response and high availability.

6. Data Consumption

- Used by traffic authorities, commuters, and emergency services.
- Supports congestion analysis and route optimization.
- **Challenge:** Accurate and up-to-date information.

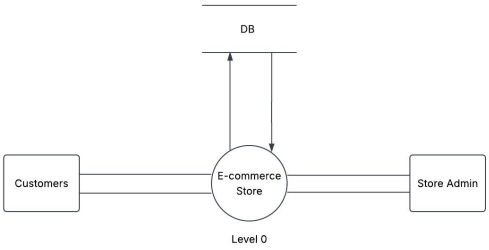
7. Monitoring & Maintenance

- Monitor sensor health and pipeline performance.
- Detect failures and data anomalies.
- **Challenge:** Ensuring continuous operation and data quality.

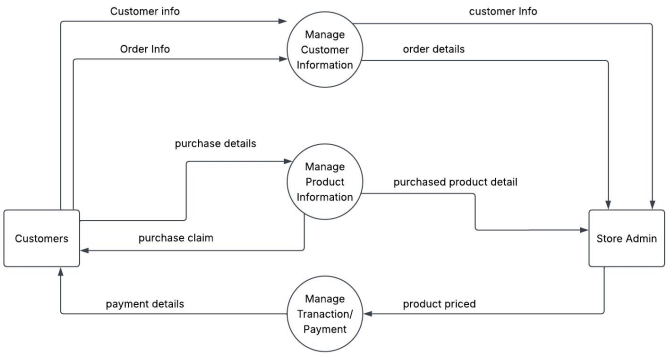
Major Challenges of Continuous Real-Time Streaming

- High-speed continuous data generation.
- Low-latency processing.
- Handling network failures and data loss.
- Scalable storage and processing.
- Ensuring data security and privacy.
- Continuous monitoring and observability.

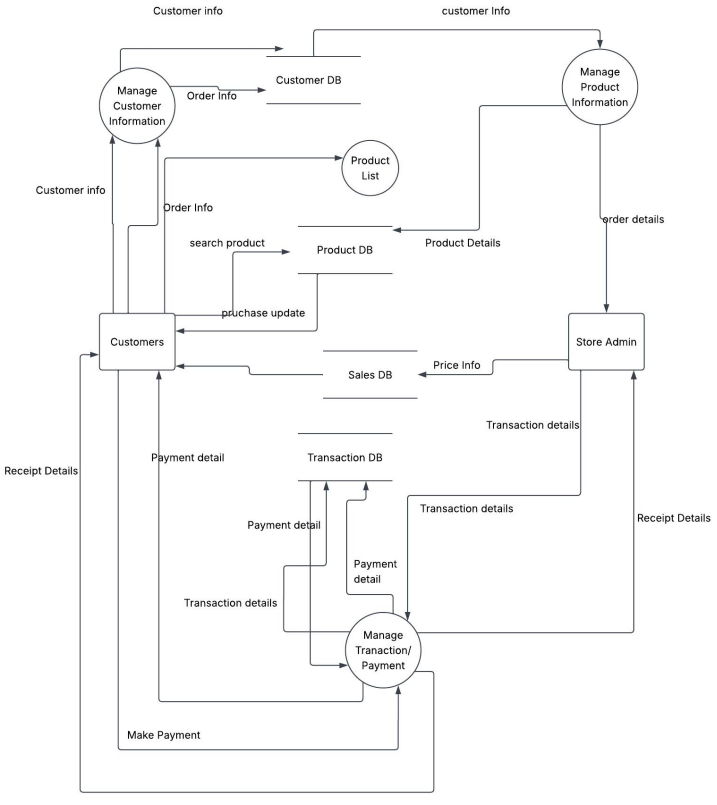
OUTPUT:



Level 0



level 1



Level 2